Session 13 — Impairments II: Case Study, Biosensors International Group

By the end of this session you must be able to apply Session 12's IAS36/IAS38 framework to a real acquisition (identify what the acquired intangibles and goodwill represent, how they were measured and allocated to CGUs, and how the impairment review was performed), and be able to critically discuss why an impairment review is a highly judgemental process, using Biosensors International Group's acquisition of JW Medical Systems (JWMS) as the running example.

Plan

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1. The Case Setup: A Sudden Jump in Intangibles and Goodwill

- Biosensors International Group (BIG), a Singapore-listed biotechnology company developing, manufacturing and commercialising medical devices for interventional cardiology (drug-eluting stents, DES) and critical care, acquired the remaining 50% of JW Medical Systems Ltd (JWMS) in October 2011, having already held 50% since 2003.
- An equity analyst reviewing BIG's FY2012 balance sheet noticed that intangible assets and goodwill together jumped from 4% of total assets the year before to 62% of total assets — almost entirely explained by the JWMS acquisition.
- This is the case's opening hook: a purchase-accounting event can transform a balance sheet's composition overnight, and the analyst's job (and this session's job) is to understand what that transformation actually represents.

Scale of the balance-sheet shift (Exhibit 5, consolidated, US$'000): intangible assets rose from 2,758 (2011) to 162,008 (2012); goodwill rose from 16,536 (2011) to 622,181 (2012). Together, these two items constituted 95% of BIG's non-current assets and 70% of its net assets at 31 March 2012.

2. The JWMS Acquisition

- JWMS, based in Weihai, Shandong Province, was originally a 50:50 joint venture between BIG and Shandong Weigao Group Medical Polymer Co. Ltd, formed in 2003 to manufacture critical-care products in China.
- Its core product, the Excel DES, was the first stent in China with a biodegradable polymer coating, and JWMS grew to become the third-largest DES player in China by market share.
- On 3 October 2011, BIG acquired the remaining 50% interest from Weigao for US$478 million (cash and stock, including convertible notes), making JWMS a wholly-owned subsidiary and Weigao BIG's single largest shareholder (21.6%, upon conversion).
- BIG's reported share of JWMS's profit for FY2011 was US$19 million, meaning the US$478 million paid for the remaining 50% implied a valuation multiple of approximately 25× JWMS's FY2011 profit after tax — a useful anchor figure for discussing whether the price paid was reasonable.

3. What the Acquired Intangibles Represent, and How They Are Measured

Recognition of acquired intangibles (accounting policy note, consistent with Session 11): intangible assets acquired in a business combination are measured at fair value at the acquisition date; following initial recognition they are carried at cost less accumulated amortisation and accumulated impairment losses.

Breakdown of BIG's intangible assets acquired via the JWMS business combination (Exhibit 8, Note 7, US$'000): patents 18,825; customer relationships 135,128; land use rights 1,841 — total additions via business combination of 155,794, against total intangible assets on the balance sheet of 162,008 at 31 March 2012. Customer relationships alone represent the overwhelming majority of the intangible balance (131,236 net book value) — precisely the kind of item Session 11 flagged as *not* recognisable when internally generated, but recognisable when acquired through a business combination.

Amortisation periods applied (straight-line):

- Computer software costs — 3 years.
- Development costs — 3 years.
- Patent costs — 10 years.
- Customer relationships — 10 years.
- Land use rights — 30 and 46 years.
- Note that the land use rights acquired via the JWMS deal (US$4,932,000, a separate addition from the business-combination

line) were not yet being amortised at 31 March 2012 because the lease period only began 1 April 2012 — consistent with Session 11's rule that amortisation begins once an asset is ready for its intended use, not before.

4. Goodwill: Recognition, CGU Allocation and the Impairment Review

Goodwill recognition (accounting policy note): goodwill on acquisition is initially measured at cost, being the excess of consideration transferred (plus any non-controlling interest recognised) over the net identifiable assets acquired and liabilities assumed. After initial recognition, goodwill is measured at cost less accumulated impairment losses — never amortised, exactly as in Session 12.

Goodwill from the JWMS acquisition (Exhibit 8, Note 8, US$'000): goodwill additions via business combinations in FY2012 totalled 607,670, of which JWMS alone contributed 618,149 to the year-end CGU allocation (net of the pre-existing goodwill balances from other CGUs that shrank or were written off) — by far the largest component of BIG's total goodwill of 622,181.

- Applying Session 12's allocation rule directly: BIG allocates goodwill to the CGUs expected to benefit from the combination's synergies, and each CGU-allocated goodwill balance is tested for impairment at least annually and whenever an impairment indicator arises.
- BIG's own five goodwill-bearing CGUs at 31 March 2012 were: BSI (comprising BSI and JWICU together, as one CGU), Bio Japan, Indonesia market for BIT, Bifurcation stents, and JWMS.
- The value-in-use calculation for each CGU used a five-year discounted cash flow forecast, with a terminal growth rate applied beyond that horizon, and disclosed assumptions per CGU: average growth rate, terminal growth rate, and pre-tax discount rate (for JWMS: 18% average growth, 2% terminal growth, 12% pre-tax discount rate).

Where impairment actually was recognised: despite the enormous JWMS-related goodwill balance, BIG's FY2012 goodwill impairment loss of US$12,232,000 was attributable entirely to three *other*, much smaller CGUs — BSI, Bio Japan and Bifurcation stents — not to JWMS. JWMS's goodwill was concluded to require no impairment charge at all.

5. The Analyst's Question: Is the "No Impairment" Conclusion Reliable?

- This is the crux of the case and the natural essay/discussion question.
- The value-in-use methodology behind the "no impairment" conclusion for JWMS rests entirely on management's own assumptions — a discounted cash flow model built on a five-year budget, an 18% average growth rate, a 2% terminal growth rate and a 12% discount rate, none of which are independently verifiable from outside the company.
- IAS36 itself acknowledges this sensitivity (Session 12, Section 1's definition of value in use, and the CGU discussion): the recoverable amount is "most sensitive to the discount rate… as well as the expected cash inflows and the growth rate used for extrapolation," in the case's own words.

The case also gives the analyst an independent cross-check: market capitalisation.

- On 2 July 2012, BIG's shares closed at S$1.23, with 1,723 million shares outstanding, implying a market value of S$2,119 million (approximately US$1,670 million at the prevailing exchange rate).
- Over the prior 52 weeks the share price ranged from S$1.09 to S$1.70, implying a market-value range of roughly US$1,480 million to US$2,309 million.
- Session 12's own list of external impairment indicators includes exactly this comparison: "the book value of the firm exceeds its market capitalisation" is one of the minimum indicators a company must consider.
- Comparing this market-value range with BIG's net book value (total equity of US$1,117,667 thousand, i.e. approximately US$1.12 billion, at 31 March 2012) is the natural next step the case sets up for class discussion, even though the case itself does not draw the conclusion for the reader.

Watch out: the case narrative deliberately stops short of concluding whether BIG's "no impairment" determination for JWMS goodwill was appropriate — it hands the reader the raw inputs (assumptions, market cap range, net book value) and leaves the judgement to classroom discussion. Do not treat any single number here as "the answer"; the point of the session is to practise using IAS36's indicator checklist and CGU-testing mechanics on ambiguous, real-world facts.

6. Assignment Questions Framework

The case's own discussion questions (from the session outline) give a structured way to organise a written or oral answer, and are worth reproducing as a checklist:

Must know (how to structure a case answer):
• **Why the premium?** Tie the US$478m price (≈25× FY2011 profit) to BIG's strategic rationale (China market access, JWMS's distribution network and manufacturing base, Weigao's shareholding and complementary global product range)
• Where do unidentifiable benefits end up? Any acquired value that cannot be separately identified (Session 11's identifiability test) is not a specific intangible — it is subsumed into goodwill
• Categories of intangibles recognised: patents, customer relationships, land use rights (per Exhibit 8), each separately identifiable and fair-valued at acquisition
• Values attributed at acquisition: patents 18,825; customer relationships 135,128; land use rights 1,841 (US$'000, per Exhibit 8)
• **Accounting treatment at acquisition and subsequently:** intangibles at fair value then cost less amortisation/impairment (Session 11); goodwill at cost then cost less impairment only, never amortised (Session 12)
• **Underlying assumptions to interrogate:** the growth rate, terminal growth rate and discount rate feeding the value-in-use DCF — all management-controlled and all capable of masking an impairment that should have been recognised

7. Common Mistakes and Exam Traps

- Assuming a "no impairment" conclusion means an asset's fair value definitely exceeds its carrying amount — it only means management's value-in-use estimate, under management's own assumptions, exceeds the carrying amount; the case exists precisely to show how much judgement that involves.
- Treating customer relationships or similar items as recognisable only when generated internally — the case is a clean real-world illustration of Session 11's rule that such items are recognisable specifically because they were acquired in a business combination, not developed internally.
- Forgetting that goodwill is tested CGU by CGU, not as one lump sum for the whole company — BIG's impairment charge hit three specific, smaller CGUs while its largest goodwill balance (JWMS) was untouched, which only makes sense once you remember goodwill impairment testing is CGU-specific.
- Missing that land use rights acquired but not yet amortising (lease not yet started) is not an error or omission — it is the correct application of the "capable of operating as intended" recognition-cutoff logic from Sessions 9–11.
- Citing the acquisition multiple (≈25× profit) as proof of overpayment on its own — the case frames it as a data point for discussion (alongside strategic rationale and market comparables), not as a conclusive test under any IAS36/IFRS3 rule.

8. Must Know Absolutely

Must know:
• BIG acquired the remaining 50% of JWMS for US$478m in Oct 2011 (≈25× JWMS FY2011 profit), taking intangibles + goodwill from 4% to 62% of total assets
• Acquired intangibles (patents, customer relationships, land use rights) are recognised at fair value on acquisition, then held at cost less amortisation/impairment — customer relationships dominate the balance (131,236 of 162,008 net)
• Goodwill (622,181 total, ≈618,149 from JWMS) is never amortised, allocated to CGUs, tested annually via value-in-use DCF (5-year forecast + terminal growth, discounted)
• FY2012 goodwill impairment (US$12,232k) hit three *smaller* CGUs (BSI, Bio Japan, Bifurcation stents), not JWMS — goodwill impairment is CGU-specific, not company-wide
• Market capitalisation exceeding/falling below book value is an explicit IAS36 impairment indicator — useful as an external sanity check on management's DCF-based value-in-use conclusion

Watch Out

Watch out: figures in this fiche are drawn from the case's Exhibits 5–8 (BIG's FY2012 annual report excerpts, as reproduced in the case) and from the session outline's assignment questions; the case itself runs to 25 pages including extensive market/industry background (Exhibits 1–4) not summarised here, since it is context rather than IAS36/IAS38 mechanics. Re-read the full case before the exam if industry-context questions (e.g. DES market dynamics, competitive landscape) are also examinable.